

Automatic Vehicle Classification Using Wireless Magnetic Sensor

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Abstract – This paper proposes an extension to our previous work on an automatic low-computed vehicle classification using embedded wireless magnetic sensor. A realization of our vehicle classification on embedded wireless magnetic sensor is studied in this work. The implementation allows real-time vehicle classification based on vehicle magnetic length, averaged energy, and Hill-pattern peaks. The system automatically detects vehicles, extracts features, and classifies them. The three features are of low-computation. We classify vehicles into 4 types: motorcycle, car, pickup and van). The classification shows a promising result. It can classify motorcycle with 95% accuracy. The classification rates between 70%-80% are achieved with car, pickup and van due to their similarity in these extracted features. The results obtained are comparable to our implementation using PC in our previous work and demonstrate that the algorithm can be realized on the embedded wireless magnetic sensor.

Keywords – vehicle classification, magnetic sensor, wireless sensor network

I. INTRODUCTION

For traffic management, the vehicle classification is important in several applications. For example, automatic toll collection requires that different vehicle type is charged differently. Road construction and maintenance must be planned using statistic data of road usage as higher grade of materials must be employed when the roads must support larger trucks. To control the road access is another application of vehicle classification. In order to develop a practical classification system, some requirements have to be considered such as high accuracy of classification with low-computation, robustness to various weather conditions, and ease of installation and maintenance. Existing several classification systems are based on inductive loop sensor, magnetic sensor, video camera, ultrasonic device and infrared camera [1]. Between them, magnetic sensor is of smaller size, easier installation and maintenance, more power efficient with high sensitivity. Magnetic sensor has been used in many classification systems. Michael J. Caruso and Luckey S. Withanawasam propose the use of three-axis Anisotropic Magnetoresistive (AMR) sensor in several vehicle applications including classification [2]. Sing Y. Cheung and et.al, also propose a simple classification scheme based on AMR sensor to classify vehicle into five types as

in [3],[4]. The magnitudes of magnetic length of three axes are used for classification giving a promising classification rate. Multiple sensors which include video, infrared and piezoelectric sensor are employed as in [5]. Using multiple sets of information from various sensors may achieve better accuracy, but also increases more complexity and computation resources. In “Ref. [6]” we design a wireless magnetic sensor that provides an implementation of real-time classification system. However, we only employ one set of three-axis magnetic sensors which do not allow vehicle speed estimation. The speed estimation is therefore obtained with video data and thus the system is not self-contained. The result is very promising. We work on an extension of our previous work such that a self-contained wireless magnetic sensor is achieved.

In this paper, an enhanced self-contained system is proposed. We use magnetic sensor because of its robustness to weather, ease of installation, and low maintenance cost. We separate the classification into two parts. In the first part, the sensor board detects vehicle and computes relative vehicle length and occupancy time. The board then samples magnetic signal and transmits these samples to the PC along with the length and occupancy time information. In the second part, the PC computes the other two features i.e. averaged energy and number of Hill-pattern peaks. We employ all three features for classification. Due to time constraint in preparing this manuscript we extract averaged energy and number of Hill-pattern peaks using a PC, but their simplicities make them possible to be computed real-time by microcontroller.

In Section II, the architecture of the proposed system is shown. The detection mechanism of vehicle arrival and departure is explained in Section III. The feature extraction and classification processes are described in Section IV and Section V. Then, the experimental results are reported in Section VI. Finally, the conclusion is in Section VII.

II. OVERVIEW OF THE PROPOSED SYSTEM

In “Fig. 1”, the system architecture of vehicle classification system is shown. The proposed system consists of magnetic sensor board, RS232 transceiver board and personal computer. The CC1000 RF

communication board is not active except for transmitting data via wireless network. On magnetic sensor board, MSP430F449 is the main processor which has many peripherals embedded inside such as A/D converter, timer module, UART and digital I/O port. Two 12-bit-ADCs convert analog signal from magnetic sensor to 4 bytes digital data. Timer module sets a sampling period to approximately 10,000 samples per second. UART module communicates and transmits sampled magnetic signal and relative vehicle length to the PC for further operations. Digital I/O port drives LED status and controls set/reset circuit for refreshing sensitivity of magnetic sensor every 2 seconds (upon the applications).

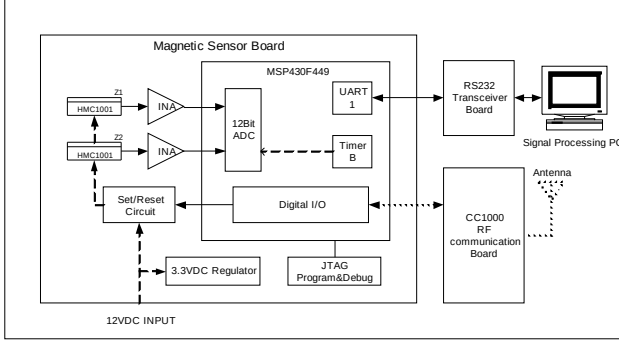


Fig. 1. System Architecture.

Two magnetoresistive sensors (HMC1001) are employed to sense density of earth magnetic field. These two sensors are separated from each other at a certain distance and allow vehicle speed estimation. The structure of sensor is nickel-iron thin film strip deposited on silicon wafer and arranged to Wheatstone bridge. By applying voltage to sensor, the output voltage can be measured. The variation of output voltage is related to a variation of the strip resistance upon magnetic flux density. This output voltage variation is too small. An amplifier is needed to magnify the signal. Two cascades amplifiers with total approximately gain of 100 are used for this purpose. The first one is instrument amplifier which has a common mode rejection characteristic and high impedance balanced input while the second one is the amplifier with active filter for reducing ripple of signal.

RS232 transceiver board is used for converting 3.3 logical voltage signal of magnetic sensor board to RS232 line level before transmitting to personal computer. At computer side, this transmitted magnetic signal is further analyzed in feature extraction and classification process.

III. VEHICLE DETECTION

First, vehicle detection is performed by magnetic sensor board processor. Since, the magnetic sensor board has limitation on computation and resources, only the earth magnetic field in z-axis is used for vehicle detection. In first stage, the earth magnetic fields of the sensor locations are set to be the base line b_1 and b_2 as shown in “Fig. 2”.

In normal situation, earth magnetic field variation has very small change, however, when a vehicle is present, the magnetic field will be disturbed and results in noticeable larger changes. If the magnitudes of earth magnetic field comparing to base line of both sensors are greater then the threshold th for a certain period of time T_d , it is considered as vehicle arrival. Otherwise, it is considered as a noise. Let z be z-axis magnetic time series observed by sensor.

z_n = z-axis magnetic field at time sample n ,
 b_i = magnetic base line of sensor i ,

$$|z_n - b_i| \geq th \quad (1)$$

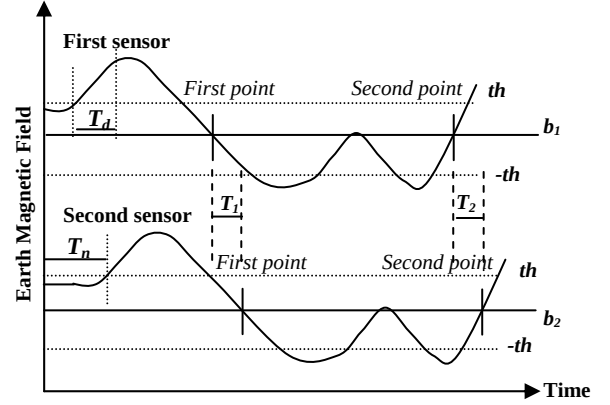


Fig. 2. Vehicle detection scheme.

When vehicle arrival is detected, the follow z-axis magnetic signal is gathered and smoothed. To smooth the signal in time series, we perform moving average over 8 samples:

$$z_n^M = \frac{\sum_{i=0}^7 z_{n-i}}{8} \quad (2)$$

Where z_n^M is moving average z-axis at time sample n . Superscript M represents a moving average sample.

Every 10 milliseconds (ms) of moving average signal, z_n^M is sent to personal computer through RS232 transceiver board. The signal transmission is stopped when vehicle departure is detected. The vehicle departure is determined as same as the arrival except the magnitudes of earth magnetic field comparing to base line of both sensors are less than the threshold for a certain period of time T_d . Otherwise, it is considered that the vehicle still occupies the sensor.

$$|z_n - b_i| \leq th \quad (3)$$

IV. FEATURE EXTRACTIONS

Features using for classification are relative vehicle length, averaged energy and numbers of Hill-pattern peak. Relative vehicle length is obtained during vehicle detection by magnetic sensor board. Averaged energy and numbers of Hill-pattern peak are obtained by personal computer.

A. Relative Vehicle Length *(please correct this part)*

Two sensors are installed on the magnetic sensor board with an inch apart. The signal delay between two sensors is used to calculate the relative speed of a vehicle. Let d be the distance between the two sensors in inch, N be the number of delay samples and v be the estimated vehicle speed.

$$v = N / d \quad (4)$$

The number of delay samples N is obtained based on the two-stage scheme. In First stage, when the vehicle arrival is detected by one sensor, the counter C is reset and increments at every sample it monitors. The increment is stopped if the second sensor detects the arrival of that vehicle. The value of counter C represents the delay sample D_s of two sensors within the delay period T_n in “Fig. 2”.

In second stage, we further explore the entire samples from arrival to departure to find the zero-crossing points. The zero-crossing point is the point where the signal crosses the base line of the sensor between the threshold boundaries. The crossing point inside the threshold boundaries is not considered as zero-crossing point. It is possible to have more than one zero-crossing point as in “Fig. 2”. If the zero-crossing point is found, the number of delay samples N is obtained as in (5). Otherwise, the number of delay samples N is set to D_s .

$$N = \frac{\sum_{i=1}^{N_z} S_{T_i}}{N_z} \quad (5)$$

Where T_i is the zero-crossing delay period of two sensors at i point.

S_{T_i} is the number of samples in period T_i .

N_z is the number of zero-crossing point.

After the estimate vehicle speed v is obtained, relative length l in inch can be calculated. Let M is the total number of samples from arrival to departure.

$$l = M / v \quad (6)$$

B. Averaged Energy

Averaged vehicle energy can be employed for

classification. It gives information on how much effect a vehicle has on the earth magnetic field at the sensor node. It is expected that large body vehicles has more effect than small body vehicles. Also, the closer the body to the sensor node also yields more changes of the signal. Averaged vehicle energy E is defined as the total energy of smoothed differential magnitude M_n^m , defined as in [6], computed during the period over which a vehicle is detected, divided by the vehicle magnetic length S

$$E = \frac{\sum_{\text{Vehicle Detected Period}} M_n^m}{S} \quad (7)$$

C. Number of Hill-pattern peak

Hill pattern is used in [3]. It is a pattern extracted from magnitude signal based on its slope. Hill-pattern is employed as it is believed that different vehicles possess different parts of structures and different numbers of axles. This is represented as the number of peaks in the Hill-pattern. We represent positive slope with 1 and negative slope with 0. A peak is considered valid when a consecutive number of samples are larger than a threshold.

V. CLASSIFICATION

We perform classification based on three features, relative vehicle length, averaged energy and number of Hill-pattern peaks into 4 types of vehicles: motorcycle, car, pickup and van. We propose a simple classification based on hierarchical tree methodology as in [6]. In “Fig. 3”, the motorcycles’ relative length, averaged energy, and number of Hill-pattern peaks are much lower than the other types of vehicles. These three features are employed in path 1 to mainly identify motorcycles and cars. A more thorough look into the groups of cars, pickup, and vans is then considered based on combination of the three features by path 2.

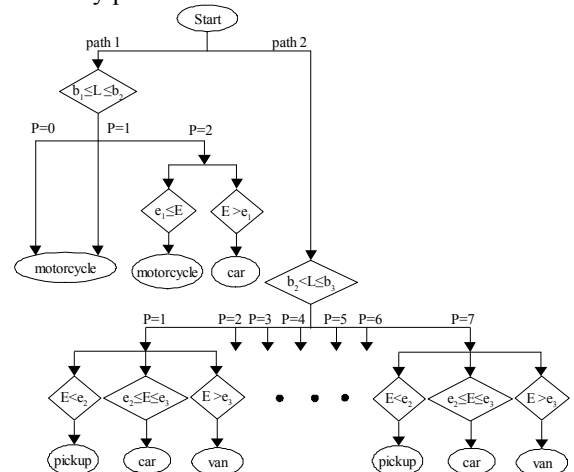


Fig. 3. Classification tree based on the three features: relative vehicle length, averaged energy, and Hill-pattern peaks.

VI. RESULTS

We setup the system by placing our magnetic sensor board on the middle of the lane about 5 meters behind the entering gate where a vehicle must enter with low speed. This allows us to collect magnetic signal with not much speed variation. With the sensors on magnetic sensor board, we obtain the earth magnetic field variation of both sensors in z-axis. Z-axis is the direction perpendicular to the ground pointing upward. The total number of vehicles we observed is 130. That consists of 24 motorcycles, 57 cars, 35 pickups and 14 vans. We do the data analysis on observed vehicles as follows.

A. Distribution of Relative Vehicle Length

“Fig. 4” shows the distribution of relative vehicle length of all 4 types of observed vehicles. We observe that the distribution of motorcycle length is clearly separated from the others types while the distributions of car, pickup and van are very much overlapped. It means that only relative vehicle length can be used to clearly classify motorcycles. However, in order to classify car, pickup and van, additional features are required.

B. Distribution of Averaged Energy

“Fig. 5” shows the distribution of averaged energy of 4 types of vehicles. Due to the size of the motorcycles, their averaged energy should be smaller than others but the distribution of motorcycles in “Fig. 5” is not much different from the others. This is because the sensors can detect the motorcycles only when they pass by very closer to the sensors that makes high earth magnetic field variations. This feature is not useful for classifying motorcycle from other types.

In “Fig. 6”, we focus on the distribution of car, pickup and van. The graph shows that the van’s average energy distribution is relatively higher than car and pickup. We can take advantage of this feature to classify van from car and pickup. The distributions of car and pickup are almost the same.

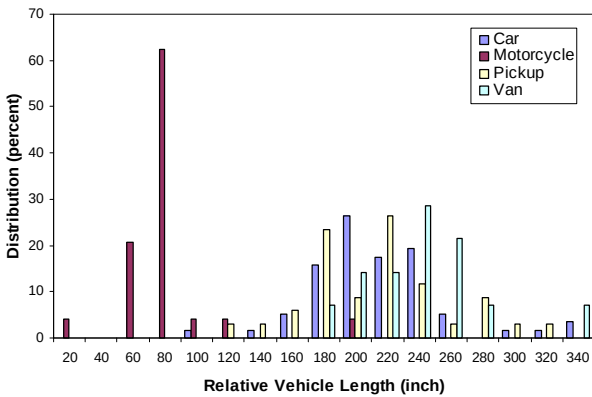


Fig. 4. Distribution of relative length.

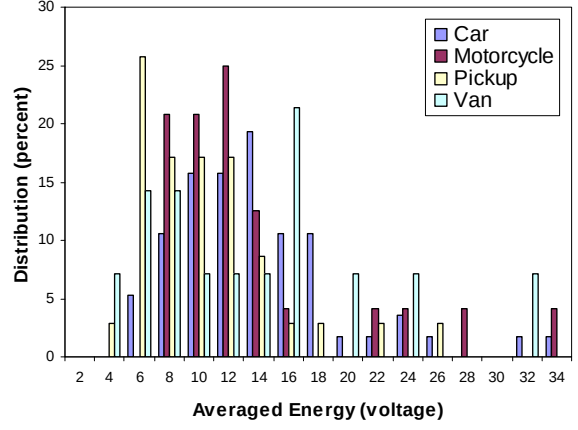


Fig. 5. Distribution of averaged energy for all types of vehicles.

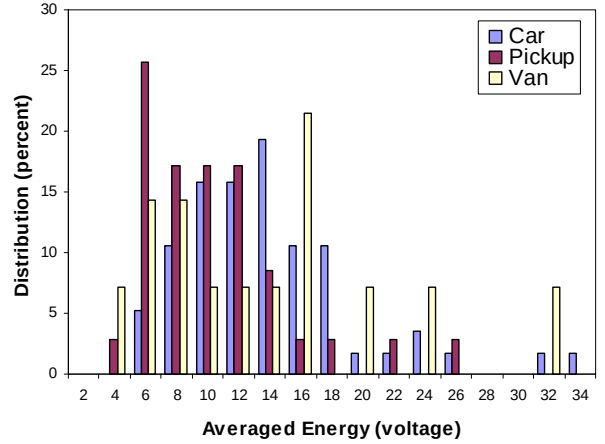


Fig. 6. Distribution of averaged energy for car, pickup and van.

C. Distribution of Number of Hill-pattern Peaks

“Fig. 7” shows that the distributions of number of Hill-pattern peaks for 4 types of vehicles. The graph shows that the motorcycle yields small number of peaks comparing to other types. For other three types, they possess almost all number of peaks, especially between 1 and 6.

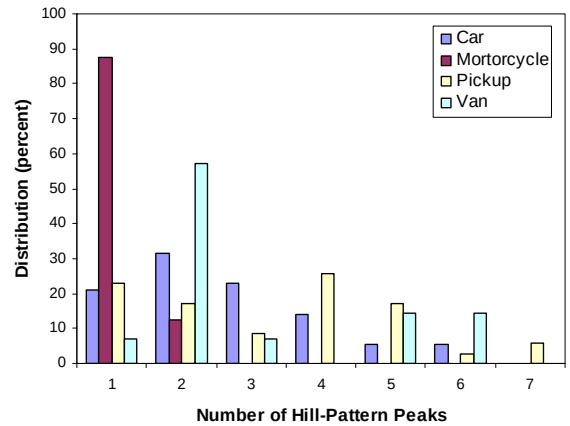


Fig. 7. Distribution of Number of Hill-Pattern Peaks.

Based on above distributions, the relative vehicle length is mainly used to classify motorcycle from other types of vehicles. All three features are used to classify car, pickup and van. The classification process is based on hierarchical tree shown in “Fig. 3”. The thresholds are chosen as follows: $b_1=3$, $b_2=110$, $b_3=300$, $e_1=8$, $e_2 \sim 6-10$, $e_3 \sim 12-16$. Thresholds e_2 and e_3 are expressed in ranges because they have to be chosen according to each different path. The overall 77.69 percent of the vehicles are classified correctly. In Table 1, the classification results of each vehicle type is shown. The motorcycle achieves high accuracy 95 percent because the length is clearly distinguished to the other three types. The discrimination of three types (car, pickup and van) is fuzzy using only the length, therefore the further classification are needed using number of Hill-pattern peak and averaged energy. The accuracies of car, pickup and van are 68.42, 95.83 and 71.43 percents respectively.

Table 1
CLASSIFICATION RESULTS

Vehicle Type	Classification Result (percent)
Car	68.42
Motorcycle	95.83
Pickup	82.86
Van	71.43

VII. CONCLUSION

An automatic vehicle classification system based on magnetic sensors for wireless sensor network is proposed. All processes of the proposed system such as data acquisition, data preprocessing, feature extraction and classification are automatically performed. The system consists of magnetic sensor board, RS232 transceiver board and personal computer. The magnetic sensor board can detect vehicle and computes its relative length. It then samples signal and transmits the signal and relative vehicle length to the PC through RS232 transceiver board.

The other two features, averaged energy and numbers of Hill-pattern peak and classification process are operated on the PC. The feature extraction and classification processes in personal computer can be further implemented on magnetic sensor node for real-time operations. All three features, relative vehicle length, averaged energy and number of Hill-pattern peaks are used for classification process. Based on the results, the system classifies vehicles into four types: motorcycle, car, pickup and van. The accuracies of car, pickup and van are 68.42, 82.86 and 71.43 percents, respectively while the accuracy of motorcycle is 95.83 percent. We found that the classification based on the length of the vehicle gives very high accuracies when the vehicles have length distinction (motorcycle). The discrimination of three types (car, pickup and van) is fuzzy using only the length, therefore the further classification are needed using peak

and averaged energy. To better separate car, pickup and van, the investigation for suitable features is needed. However, the feature extraction processes for these features must be simple enough to implement on magnetic sensor board.

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